

Dynamic DAGs for the Contemporaneous Treatment Effect

Technical note for the IVB paper

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1 Purpose and conventions

This note documents the three time-indexed causal DAGs embedded as reproducible TikZ in `ivb_paper_pa.Rmd`. Their target is the contemporaneous treatment effect (CET),

$$\tau_{\text{CET}}(d, d'; h) = \mathbb{E}[Y_t(d) - Y_t(d') \mid H_t^- = h],$$

where the opening history H_t^- is fixed before the first exposure represented by D_t , and Y_t is measured after the relevant within-period responses. Unless a controlled direct effect is stated, the target is the total CET. The Task 03 sequence $H_t^- \rightarrow D_t \rightarrow Z_t \rightarrow Y_t$ is the candidate clock for the mediator mechanism, not a universal order. The required collider $D_t \rightarrow Z_t \leftarrow Y_t$ instead places the outcome process represented by Y_t before the measurement of Z_t within the period. The lagged variables shown in a graph belong to H_t^- only if their substantive timestamps pass the Task 03 checklist. The graphs do not assert that a calendar lag is automatically pretreatment, that the displayed history is sufficient, or that a within estimator is unbiased at finite T .

A path is evaluated using the usual d-separation rules. An unconditioned noncollider transmits association; conditioning on a noncollider blocks the path. An unconditioned collider blocks a path; conditioning on that collider or one of its descendants opens the path. These path statements concern the displayed DAG and do not rule out omitted common causes, interference, simultaneity, misspecified lag order, or treatment-effect heterogeneity.

2 Graph 1: contemporaneous collider with an inherited lag path

2.1 Edges and target

The graph contains the causal edges $D_{t-1} \rightarrow Y_{t-1}$ and $D_t \rightarrow Y_t$, the contemporaneous collider $D_t \rightarrow Z_t \leftarrow Y_t$, persistence $D_{t-1} \rightarrow D_t$, $Y_{t-1} \rightarrow Y_t$, and $Z_{t-1} \rightarrow Z_t$, and the lagged collider $D_{t-1} \rightarrow Z_{t-1} \leftarrow Y_{t-1}$. The target is the current total CET along $D_t \rightarrow Y_t$.

2.2 Open and closed paths

- With neither Z_t nor Z_{t-1} conditioned on, $D_t \rightarrow Z_t \leftarrow Y_t$ is closed at Z_t , and $D_t \leftarrow D_{t-1} \rightarrow Z_{t-1} \leftarrow Y_{t-1} \rightarrow Y_t$ is closed at Z_{t-1} .
- Conditioning on Z_t opens the contemporaneous noncausal path $D_t \rightarrow Z_t \leftarrow Y_t$.
- Because Z_t is a descendant of Z_{t-1} , conditioning on Z_t also opens the inherited path through the lagged collider: $D_t \leftarrow D_{t-1} \rightarrow Z_{t-1} \leftarrow Y_{t-1} \rightarrow Y_t$.

- Independently of collider conditioning, $D_t \leftarrow D_{t-1} \rightarrow Y_{t-1} \rightarrow Y_t$ is an open ordinary dynamic backdoor path unless the relevant lagged treatment or outcome history is conditioned on.
- Conditioning on Y_{t-1} blocks both cited lag paths because Y_{t-1} is a noncollider on each. It does not block the contemporaneous path opened by conditioning on Z_t .

2.3 Conditioning set

For the displayed pure-collider graph, the minimal set for the collider mechanisms excludes both Z_t and Z_{t-1} . The contemporaneous Z_t remains inadmissible for the total CET because it opens $D_t \rightarrow Z_t \leftarrow Y_t$. The lag Z_{t-1} is not intrinsically inadmissible under every joint set: if it is needed for another documented path, conditioning jointly on Y_{t-1} blocks the inherited path that conditioning on Z_{t-1} opens. Lagged outcomes or treatments may likewise belong to H_t^- for assignment or dynamic-mean reasons, subject to the Task 04 conditions. Adding Y_{t-1} does not cure conditioning on Z_t .

3 Graph 2: dual-role control

3.1 Edges and target

At period t , Z_{t-1} is a joint cause of treatment and outcome: $Z_{t-1} \rightarrow D_t$ and $Z_{t-1} \rightarrow Y_t$. It is therefore a confounder for the CET if its timestamp is genuinely pretreatment. The current variable is a collider, $D_t \rightarrow Z_t \leftarrow Y_t$, with $Z_{t-1} \rightarrow Z_t$. The lagged state also preserves the preceding-period collider $D_{t-1} \rightarrow Z_{t-1} \leftarrow Y_{t-1}$, the preceding-period causal edge $D_{t-1} \rightarrow Y_{t-1}$, treatment persistence $D_{t-1} \rightarrow D_t$, and outcome persistence $Y_{t-1} \rightarrow Y_t$. The target remains the total CET carried by $D_t \rightarrow Y_t$.

3.2 Open and closed paths

- Without conditioning, the backdoor path $D_t \leftarrow Z_{t-1} \rightarrow Y_t$ is open. The inherited path $D_t \leftarrow D_{t-1} \rightarrow Z_{t-1} \leftarrow Y_{t-1} \rightarrow Y_t$ is closed at the collider Z_{t-1} .
- Conditioning on Z_{t-1} blocks the confounding fork because Z_{t-1} is a noncollider on $D_t \leftarrow Z_{t-1} \rightarrow Y_t$.
- The same conditioning opens the inherited path because Z_{t-1} is a collider on $D_t \leftarrow D_{t-1} \rightarrow Z_{t-1} \leftarrow Y_{t-1} \rightarrow Y_t$.
- The distinct path $D_t \leftarrow D_{t-1} \rightarrow Y_{t-1} \rightarrow Y_t$ remains open without lag adjustment and does not depend on conditioning at Z_{t-1} .
- Conditioning additionally on Y_{t-1} blocks both the inherited collider path and the ordinary dynamic path at Y_{t-1} , which is a noncollider on each. The simulation comparison cannot decompose the improvement between those two closures.
- Conditioning on Z_t would separately open $D_t \rightarrow Z_t \leftarrow Y_t$. The set $\{Z_{t-1}, Y_{t-1}\}$ does not close that contemporaneous path if Z_t is included.

3.3 Conditioning set and scope

For the displayed dual-role graph, $\{Z_{t-1}, Y_{t-1}\}$ blocks the lagged-confounding path, the cited inherited path, and the displayed ordinary dynamic path while leaving $D_t \rightarrow Y_t$ open. This is not a universal sufficient adjustment set. Its interpretation requires that both lags are fixed before D_t , that no other backdoor path remains, that the dynamic conditional mean is correctly specified, and

that the remaining Task 04 conditions hold. It is a population d-separation claim, not a statement that the conventional within estimator is unbiased at finite T .

4 Graph 3: contemporaneous mediator with lagged confounding and state

4.1 Edges and target

The graph contains direct and mediated effects in both displayed periods: $D_s \rightarrow Y_s$ and $D_s \rightarrow Z_s \rightarrow Y_s$ for $s \in \{t-1, t\}$. The opening state Z_{t-1} also affects D_t , Z_t , and Y_t , while $D_{t-1} \rightarrow D_t$ and $Y_{t-1} \rightarrow Y_t$ encode persistence. The target is the current total CET, so both current causal paths belong to the estimand.

4.2 Open and closed paths

- Before adjustment, $D_t \leftarrow Z_{t-1} \rightarrow Y_t$ is an open backdoor path. A longer path through prior treatment, $D_t \leftarrow D_{t-1} \rightarrow Z_{t-1} \rightarrow Y_t$, is also open unless its noncolliders are conditioned on.
- The ordinary dynamic path $D_t \leftarrow D_{t-1} \rightarrow Y_{t-1} \rightarrow Y_t$ is also open before lag adjustment.
- Conditioning on a genuinely pretreatment Z_{t-1} blocks the first two backdoor paths at Z_{t-1} and does not block $D_t \rightarrow Z_t \rightarrow Y_t$. It does not by itself block the path through Y_{t-1} ; that requires D_{t-1} , Y_{t-1} , or another sufficient history representation.
- Conditioning on Z_t blocks the indirect causal path because Z_t is a noncollider on $D_t \rightarrow Z_t \rightarrow Y_t$. The coefficient on D_t then targets a direct rather than total effect only under the additional assumptions needed for mediation analysis.
- Conditioning on Y_{t-1} can be appropriate when it is part of the sufficient opening history or dynamic conditional mean, but it is not what distinguishes the total from the direct CET in this graph.

4.3 Conditioning set

For the displayed graph and the total CET, the candidate lagged adjustment set includes Z_{t-1} and any additional pretreatment history required for assignment and dynamics, such as D_{t-1} or Y_{t-1} . It excludes Z_t . This recommendation is conditional on defended timestamps, no anticipation, no unobserved contemporaneous confounding, and a correctly specified history; lagging does not by itself guarantee identification.

5 Manuscript mapping and reproducibility

The collider and mediator DAGs replace the elementary DAG figure in the timing section of `ivb_paper_pa.Rmd`. The dual-role DAG appears immediately before the existing dual-role simulation figure. All three diagrams are defined directly in the manuscript with TikZ; no external image or simulation is required to reproduce them. The legacy label `fig:three_structures_pa` is retained on the dynamic collider figure, and the new labels are `fig:dynamic_collider_pa`, `fig:dynamic_mediator_pa`, and `fig:dynamic_dual_role_pa`.

No simulation was run to produce or validate these graphs. Their arrows were matched to the mechanisms in the existing dual-role and over-control data-generating processes, while the causal

interpretation was constrained by the Task 03 timing checklist and Task 04 CET identification conditions.